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Wavelet Transform

Motivation and Key Results

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The Fourier Transform

- ▶ Perhaps the oldest known signal analysis tool

- ▶ Analysis: $X[k] = \langle x[n], e^{j2\pi kn/N} \rangle = \sum_{n=0}^{N-1} x[n] W_N^{nk},$

$$W_N = e^{-j2\pi/N}$$

- ▶ Synthesis: $x[n] = \frac{1}{N} \langle x[k], e^{-j2\pi kn/N} \rangle = \frac{1}{N} \sum_{k=0}^{N-1} x[k] W_N^{-nk}$
- ▶ Works very well with smooth signals

The Fourier Transform - Issue

- Poor time-frequency localization - poor representation of signals with sharp discontinuities

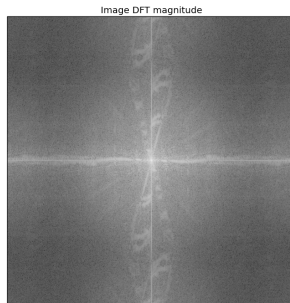


Figure: All spatial localization lost!

Why is time-frequency localization important?

- ▶ Many applications - origins in localizing earth quake origin
- ▶ Notion of scale-space also important



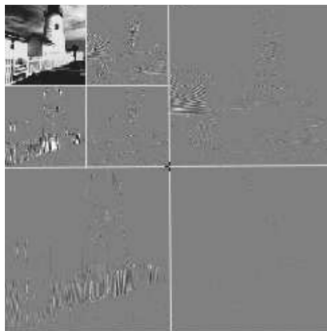
Enter Discrete Wavelets

- ▶ Defines a set of basis functions $\phi_{j,k}(x)$ and $\psi_{j,k}(x)$ that:
 - ▶ provide time-frequency localization
 - ▶ naturally afford scale-space representation
 - ▶ approximation coefficients: $F_A[j, k] = \langle f(x), \phi_{j,k}(x) \rangle$
 - ▶ wavelet coefficients: $F_W[j, k] = \langle f(x), \psi_{j,k}(x) \rangle$
 - ▶ both analysis and synthesis achieved using digital filters

Example Wavelet Decomposition



(a) Image



(b) 2-level DWT

A Few Notes

- ▶ Closed form expressions for $\phi(x)$ and $\psi(x)$ are not usually known. Notable exceptions: Haar and Gabor
- ▶ Filters derived from MRA/PR conditions are the key:
 - ▶ $\dots, V_{-1} \subset V_0 \subset V_1, \dots$
 - ▶ $V_j = V_{j-1} \oplus W_{j-1}$
 - ▶ $V_j = V_0 \oplus W_1 \oplus \dots \oplus W_{j-1}$
 - ▶ $\langle \phi(x), \phi(x-k) \rangle = \delta(k)$
 - ▶ $\langle \psi(x), \psi(x-k) \rangle = \delta(k)$
 - ▶ $\langle \phi(x), \psi(x-k) \rangle = 0$

Wavelet Analysis using the Haar Wavelet

$$\phi(x) = \begin{cases} 1, & 0 \leq x < 1, \\ 0, & \text{elsewhere.} \end{cases} \quad (1)$$

$$V_0 = \sum_{k \in \mathbb{Z}} a_k \phi(x - k), a_k \in \mathbb{R}. \quad (2)$$

$$\psi(x) = \phi(2x) - \phi(2x - 1). \quad (3)$$

$$W_0 = \sum_{k \in \mathbb{Z}} a_k \psi(x - k), a_k \in \mathbb{R}. \quad (4)$$

Wavelet Analysis using the Haar Wavelet

At a scale j , the relationship generalizes to

$$\phi(2^j x) = \frac{[\psi(2^{j-1}x) + \phi(2^{j-1}x)]}{2}, \quad (5)$$

(6)

$$\phi(2^j x - 1) = \frac{[\phi(2^{j-1}x) - \psi(2^{j-1}x)]}{2}. \quad (7)$$

Wavelet Analysis using the Haar Wavelet

$$f^j(x) = \sum_{k \in \mathbb{Z}} a_k \phi(2^j x - k) \quad (8)$$

$$= \sum_{k \in \mathbb{Z}} [a_{2k} \phi(2^j x - 2k) + a_{2k+1} \phi(2^j x - (2k + 1))] \quad (9)$$

$$= \sum_{k \in \mathbb{Z}} [a_{2k} \phi(2(2^{j-1} x - k)) + a_{2k+1} \phi(2(2^{j-1} x - k) - 1))] \quad (10)$$

$$= \sum_{k \in \mathbb{Z}} a_{2k} \left[\frac{\phi(2^{j-1} x - k) + \psi(2^{j-1} x - k)}{2} \right] + a_{2k+1} \left[\frac{\phi(2^{j-1} x - k) - \psi(2^{j-1} x - k)}{2} \right] \quad (11)$$

$$= \sum_{k \in \mathbb{Z}} \left(\frac{a_{2k} + a_{2k+1}}{2} \right) \phi(2^{j-1} x - k) + \left(\frac{a_{2k} - a_{2k+1}}{2} \right) \psi(2^{j-1} x - k) \quad (12)$$

Wavelet Analysis using the Haar Wavelet

- ▶ Analysis and synthesis done using digital filters
- ▶ DFT/IDFT done using standard basis
- ▶ $\phi(x)$ and $\psi(x)$ easily defined only for the Haar/Gabor case
- ▶ Clearly achieves space-frequency localization
- ▶ Central to several applications in image processing